

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati

Kent, OH | F-1 OPT (US work-authorized)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building and productionizing LLM applications in regulated healthcare environments. Expertise in LangChain, LlamaIndex, RAG system design, and end-to-end delivery from prototype to production-grade FastAPI/Docker services. Demonstrated in HIPAA-conscious settings: ~35% retrieval precision gains, >30% hallucination reduction, ~30% fewer post-deploy defects, audit-ready evaluation trails. Strong cross-functional communication with non-technical stakeholders. Track record of productionizing AI tools and enforcing data governance across enterprise systems. F-1 OPT - US work-authorized.

CORE COMPETENCIES

LangChain • LlamaIndex • Python • FastAPI • Flask • Docker • RAG System Design • LLM Production Engineering • Agentic Workflows • Data Governance (HIPAA) • Evaluation Pipelines • Audit-Ready Reasoning Traces • REST API • Embeddings & Vector Search • scikit-learn • XGBoost • Stakeholder Communication • Healthcare AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee | USA — Healthcare Technology, AI Consulting, SaaS

- Built **RAG systems** for clinical knowledge retrieval using recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**; retrieval-grounded alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation) with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; improved recall **15–20%** on high-risk categories at >90% precision.
- Packaged ML/LLM inference as **FastAPI/Flask** REST services containerized with **Docker**, structured logging, load simulation; reduced post-deployment defects ~**30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage docs, evaluation audit trails, and stakeholder-facing system-limitation documentation.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) | Hyderabad, India — Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures for Order-to-Cash platform; reduced query execution time ~**20%** and retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema and stored procedure deployments; reduced deployment errors >**30%** and improved release cycle time ~**35–40%**.
- Designed **Grafana dashboards** tracking long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer | Hyderabad, India — Nonprofit Tech, Video Analytics, NLP

- Built transformer-based **video summarization** system across 5,000+ sessions; reduced manual review time **60–70%** (hours → <5 min). ~85% highlight alignment with human curation.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs ~**30%**, raised contextual accuracy >**85%**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts** between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, RAG, multi-agent orchestration, FastAPI, Docker

Manga Lens — Chrome Extension for Real-Time AI Translation shipped

Full-stack browser extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider LLM orchestration with fallback routing; multi-section capture pipeline; 7-day translation cache; per-domain selector configs for 29 sites.

Manifest V3, TypeScript, Claude API, OpenAI API, Ollama (local models)

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system with an **LLM-based agent orchestrator** decomposing high-level prompts into generation tasks. Stable Diffusion fine-tuning + LoRA for identity consistency; ControlNet + OpenPose for pose conditioning.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, LLM agent orchestration

Driver Drowsiness Detection System — Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with unified **YOLOv8** model; reduced inference latency **~30%**. Sliding-window confidence aggregation for blink suppression; **~25% false positive reduction**; real-time OpenCV capture.

YOLOv8, OpenCV, Python, real-time video analytics

EDUCATION

Master's Degree — Kent State University, Ohio, USA

Aug 2023 – May 2025

CERTIFICATIONS

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL), TypeScript, React

LLM & GenAI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, evaluation pipelines, guardrails, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers

Computer Vision: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, MLOps/LLMOps, observability

Domains: Healthcare AI (HIPAA-conscious), Enterprise AI, Workflow Automation, Computer Vision